

YIWU, China

WMO: 521180

Lat: 43.2422N

Lon: 94.6724E

Elev: 5673

StdP: 11.92

Time Zone: 8.00 (E08)

Period: 94-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.4	-2.9	-23.3	1.9	0.5	-19.6	2.3	5.9	29.0	24.6	24.4	10.5	5.3	270	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	81.5	57.4	79.6	56.5	77.4	55.3	60.9	74.9	59.1	74.4	57.6	72.9	6.4	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.6	81.5	67.4	53.2	74.4	66.4	51.0	68.6	64.7	29.9	74.7	28.5	74.7	27.4	72.6	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	25.7	21.3	17.9	DB	-11.7	85.5	4.1	1.7	-14.7	86.8	-17.1	87.8	-19.4	88.8	-22.4	90.1
(5)				WB	-12.4	65.4	4.0	4.9	-15.2	68.9	-17.6	71.8	-19.8	74.5	-22.7	78.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	40.8	12.3	18.9	27.6	42.5	55.4	62.8	66.3	64.1	54.6	40.0	27.6	16.2	(6)
(7)		DBStd	20.43	7.36	6.92	10.09	8.15	7.32	5.49	4.27	5.21	6.90	8.48	9.18	8.32	(7)
(8)		HDD50	5083	1170	870	695	244	34	1	0	2	29	318	673	1047	(8)
(9)		HDD65	9007	1635	1290	1160	676	305	103	36	77	317	774	1122	1512	(9)
(10)		CDD50	1723	0	0	0	18	202	383	505	437	168	10	0	0	(10)
(11)		CDD65	173	0	0	0	0	8	36	76	47	6	0	0	0	(11)
(12)		CDH74	1407	0	0	0	4	84	286	548	423	62	0	0	0	(12)
(13)		CDH80	131	0	0	0	0	3	25	61	38	4	0	0	0	(13)
(14)	Wind	WSAvg	7.3	5.7	6.1	7.4	9.6	10.0	8.6	7.6	7.1	7.4	6.7	6.1	5.7	(14)
(15)	Precipitation	PrecAvg	3.5	0.0	0.0	0.1	0.2	0.3	0.7	1.1	0.6	0.3	0.1	0.0	0.0	(15)
(16)		PrecMax	6.0	0.1	0.1	0.7	1.1	1.1	2.2	3.1	1.1	1.1	0.4	0.3	0.2	(16)
(17)		PrecMin	1.4	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	1.2	0.0	0.0	0.1	0.2	0.3	0.4	0.7	0.4	0.3	0.1	0.0	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.9	41.9	60.0	72.8	80.0	82.8	84.6	83.4	79.2	66.2	54.9	40.7	(19)
MCWB			29.3	29.2	39.6	46.9	52.0	56.5	60.1	58.1	53.7	45.7	37.8	29.3	(20)	
2%		DB	32.0	37.2	52.1	64.3	76.4	79.9	81.6	80.9	75.2	61.4	48.7	35.7	(21)	
		MCWB	24.2	26.7	36.6	43.5	50.0	54.9	59.5	57.0	51.0	43.0	34.6	26.3	(22)	
5%		DB	27.4	34.0	48.0	60.2	73.1	77.1	79.5	78.8	71.4	57.7	44.9	32.1	(23)	
		MCWB	21.1	24.6	33.5	41.6	48.6	53.7	58.5	55.9	50.1	41.0	32.2	24.0	(24)	
10%		DB	23.6	30.8	44.0	56.7	69.4	74.9	77.3	76.2	67.8	54.6	41.1	28.3	(25)	
		MCWB	18.3	22.8	31.7	40.0	47.8	53.3	57.8	54.9	48.5	39.9	30.6	21.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.4	29.9	40.5	47.4	54.0	59.3	64.3	62.6	55.5	48.1	38.4	29.6	(27)
MCD B			37.6	39.7	56.4	69.5	76.0	77.0	79.0	76.1	73.8	63.9	53.8	39.7	(28)	
2%		WB	24.5	27.4	37.3	44.4	51.9	57.1	62.0	59.2	53.4	43.9	34.9	26.8	(29)	
		MCD B	31.3	36.0	50.5	61.2	72.6	73.5	76.4	74.3	70.7	58.7	47.8	35.2	(30)	
5%		WB	21.3	25.4	34.1	42.7	50.2	55.8	60.3	57.7	51.2	41.8	32.9	24.1	(31)	
		MCD B	26.9	33.6	46.9	57.8	69.9	72.2	74.8	73.8	68.7	55.7	43.9	31.5	(32)	
10%		WB	18.3	22.9	31.8	40.8	48.3	54.8	58.9	56.6	49.3	40.1	30.9	21.6	(33)	
		MCD B	23.4	30.7	43.4	54.4	66.6	71.2	74.1	72.6	66.0	53.3	40.8	27.9	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.9	19.0	19.6	18.4	18.8	18.0	17.9	19.3	20.2	18.7	17.6	17.8	(35)
MCD BR			20.7	21.1	23.4	23.3	23.1	21.5	21.2	22.8	23.6	22.2	20.8	20.9	(36)	
MCWBR			15.1	14.1	13.4	11.3	10.9	10.5	10.6	9.8	10.5	11.8	12.4	14.5	(37)	
5% WB		MCD BR	20.3	19.3	21.0	18.8	18.8	17.5	18.4	19.3	20.7	20.8	19.8	20.2	(38)	
		MCWBR	15.0	13.0	12.5	9.9	9.7	9.2	9.8	8.9	9.3	11.3	12.0	14.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.276	0.316	0.360	0.407	0.418	0.430	0.424	0.395	0.364	0.337	0.299	0.274	(40)	
taud		2.290	2.203	2.089	2.022	2.050	2.097	2.155	2.226	2.253	2.238	2.290	2.304	(41)		
Ebn at Noon		272	276	276	271	271	268	268	272	272	264	259	261	(42)		
Edh at Noon		27	35	45	52	52	50	47	42	38	33	27	24	(43)		
(44)	All-Sky Solar Radiation	RadAvg	710	1050	1507	1891	2201	2183	2139	1979	1680	1230	797	608	(44)	
(45)		RadStd	39	54	66	105	109	143	101	79	66	43	47	45	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+0.83	N/A	N/A	+1.17	-1.84	-2.63	N/A	N/A	+263	+182

Nomenclature: See separate page